

# Arturo Abraham López Pérez

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## EDUCATION

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### Senior Data Scientist

*Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)*  
2023-2026

### Data Science (MSc specialization)

*Institute of Executive Education and Technology (IEXE)*  
2021-2023

### MSc in Mathematics

*Benemérita Universidad Autónoma de Puebla (BUAP)*  
2006-2008

### BSc in Mathematics

*Benemérita Universidad Autónoma de Puebla (BUAP)*  
2005

## Experience

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### Digitalization and Data Governance

La Ferme — 2025

- Currently leading the implementation of a data governance framework (DCAM) at La Ferme. The project is focused on enhancing operational efficiency and product traceability and is in the development phase.
- Designed a relational data model that standardizes inventory, production, and sales management.
- Built the database architecture (PostgreSQL) to support the digitalization of key processes.
- Developed a client acquisition process using the Google Maps API to identify and manage potential clients within a defined geographic radius.
- Upcoming stages include conducting a historical data quality analysis and developing a change management plan for the operational team.

**More details:** D&G\_project — **Code:** Digitalization-Data-Governance

### DeepRL RecSys

Certification Project — 2025-2026

- Designed and developed a production-grade platform comparing collaborative filtering baselines (RecBole: Pop, BPR, NeuMF, LightGCN) against contextual bandit agents (LinUCB, Epsilon-Greedy) using the Open Bandit Dataset, with full reproducibility across 10,000+ samples and deterministic seed management.
- Implemented a comprehensive Off-Policy Evaluation (OPE) suite leveraging IPS, SNIPS, and Doubly Robust estimators with bootstrap confidence intervals, weight diagnostics (ESS, clipping analysis), and sensitivity testing—enabling reliable offline policy comparison without costly A/B tests in production environments.
- Built an interactive Streamlit dashboard providing end-to-end visibility into model performance, OPE results, and weight diagnostics, with automated fallback mechanisms and 4 functional analysis sections for stakeholder-level insights.

- Established enterprise-grade development practices: implemented automated testing (18+ pytest suite), strict type checking (mypy with `disallow_untyped_defs`), code formatting (black, ruff), CI/CD pipelines with GitHub Actions, and comprehensive CRISP-DM documentation for audit compliance and knowledge transfer.
- Validated mathematical soundness through sanity checks confirming  $IPS(\text{RealizedAction}) \approx \text{ground-truth on-policy value}$ , documented all design trade-offs (TF-Agents/Keras incompatibility workarounds with pure NumPy implementations), and published model cards with reproducible results and confidence bounds.

**Code:** DeepRL-RecSys-Platform

### **Lecturer in Mathematics and Data Science**

Universidad de las Américas (UDLAP)

2016 – 2024

- Instructed courses in key areas such as Applied Mathematics, Statistics, and Data Science.
- Designed and developed curriculum and instructional materials for hands-on sessions, focusing on the real-world application of mathematical models and Python programming.
- Delivered workshops on statistical inference, data analysis, and machine learning, training students in the use of industry-standard tools.
- Mentored student projects to help them apply theoretical knowledge to practical case studies.

**Course Details & Materials:** MDD, RDP

## **TECHNICAL SKILLS**

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### **Programming & Databases**

- **Languages:** Python, R
- **Databases:** SQL, MongoDB, Firebase
- **SQL Warehouses:** Snowflake
- **Vector Databases:** ChromaDB

### **Machine Learning & Deep Learning**

- **Frameworks:** Scikit-learn, TensorFlow, Keras, Hugging Face
- **Algorithms:** CNN, RNN, GM, DQNN

### **Big Data & Distributed Computing**

- **Frameworks:** Apache Spark, Databricks

### **Data Visualization & Business Intelligence (BI)**

- **BI Tools:** Power BI, Streamlit
- **Python Libraries:** Matplotlib, Plotly, Seaborn

### **DevOps & MLOps**

- **Version Control:** Git, GitHub
- **Containers & Orchestration:** Docker, Kubernetes
- **Workflow Orchestration:** Apache Airflow
- **Cloud Platforms:** AWS
- **ML Lifecycle Management:** MLflow, Kubeflow

# CERTIFICACIONES

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## **Senior Data Scientist (Certification)**

*Instituto Tecnológico de Monterrey (ITESM)*

2025

- Completed 70 hours of knowledge and practical application.
- **View Certificate:** [Link](#)

## **Kaggle Certifications (Online)**

Kaggle

2022

- Completed advanced specialization modules:
  - **Programming & Databases:** Python, Advanced SQL
  - **Machine Learning (ML):** Data Visualization, Deep Learning, Natural Language Processing (NLP)
  - **Reinforcement Learning (RL):** Reinforcement Learning
- **View Certificate:** [Link](#)

## **TIB2018 (Genomics Sciences Center, UNAM)**

*Universidad Nacional Autónoma de México (UNAM) — Mexico*

2018

- Completed a workshop on R and R Studio, focused on the analysis of biological data.

## **Data Analysis (Certification)**

*Johns Hopkins University — Online Certification*

2013

- Completed modules on Exploratory Graphs, Clustering, Regression, and Model Checking.
- **View Certificate:** [Link](#)